

Tactual Communication of Speech

During my Ph D, I investigated two main questions to understand the communication capabilities of the skin to mediate useful information.

1. Does the skin have the capacity to transfer high-bandwidth information derived from speech?
2. Are tactual codes quickly discerned, learned, retained and scaled over time?

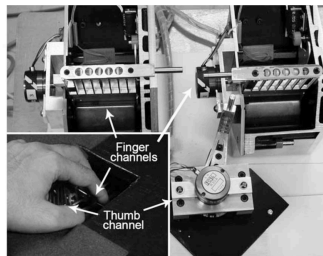
The basic idea emerged from the historical Tadoma method, where impaired individuals spread their hand on the face of a speaker and extract speech information while tracking facial actions through their hands.

I am thankful to Prof. Hong Z. Tan, Prof. Peter Meckl, Prof. Patricia Davies, and Prof. Charlotte Reed for their advice and mentoring.

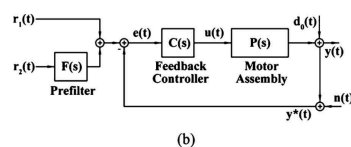


Notable Publications:

- Israr, A., Meckl, P. H., Reed, C. M., & Tan, H. Z. (2009). Controller design and consonantal contrast coding using a multi-finger tactual display. The Journal of the Acoustical Society of America, 125(6), 3925-3935. [\[Link\]](#)

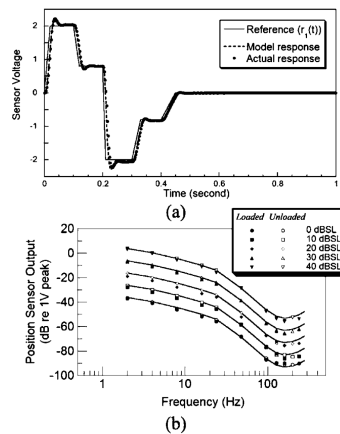


(a)

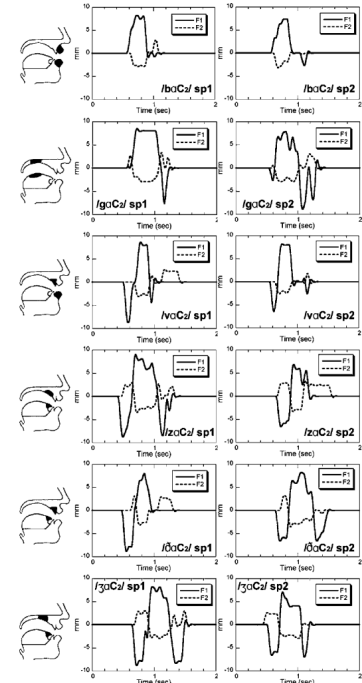


(b)

A Three-channel Tactuator device for three digits in the hand, and the structure of a 2DOF controller.



A 2DOF controller response. The low-frequency controller tracks motion with low latency and error and the high-frequency prefilter shapes the overall transfer to match that of the human sensitivity curve, thereby preserving the perceived strength of the broadband spectral information.



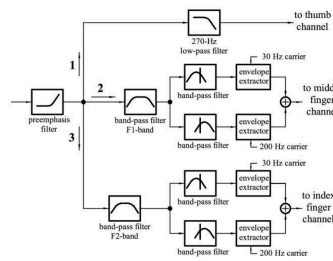
Consonantal Kinesthetic cueing for place of articulation: A realtime acoustic extraction scheme that correlates place of articulation inside the mouth to finger movements to map transitions in the oral track for speech comprehension.

- Israr, A., Tan, H. Z., & Reed, C. M. (2006). Frequency and amplitude discrimination along the kinesthetic-cutaneous continuum in the presence of masking stimuli. The Journal of the Acoustical Society of America, 120(5), 2789-2800. [\[Link\]](#)

Frequency and amplitude discrimination thresholds along the kinesthetic to cutaneous continuum were evaluated on the left index fingerpad with and without the masking stimuli. The range of frequency Weber fraction was 0.13–0.38 and 0.14–0.28, and the range of amplitude discrimination threshold was 1.82–2.98 dB and 1.65–2.71 dB, at 20 and 35 dB SL, respectively.

- Israr, A. (2007). Tactual transmission of phonetic features. Doctoral dissertation, Purdue University.

- Israr, A., Reed, C. M., & Tan, H. Z. (2008, March). Discrimination of vowels with a multi-finger tactual display. In 2008 Symposium on Haptic Interfaces for Virtual Environment and Teleoperator Systems (pp. 17-24). IEEE. [\[Link\]](#)



Vibration coding scheme from speech to tactile cues on the hand

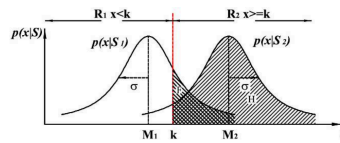
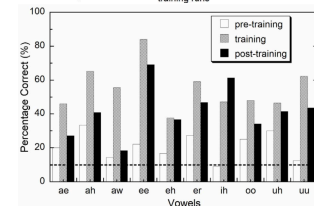
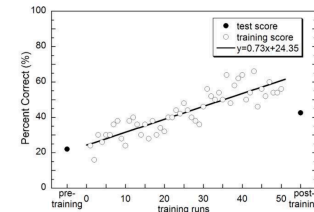


Figure 4. A decision space

The sensitivity index is defined as,

$$d' = \frac{M_2 - M_1}{\sigma}$$

The decision space for an ideal observer to differential acoustical cues extracted from vowels.



Performance scores for identification experiments:

- Wong, E. Y., Israr, A., & O'Malley, M. K. (2010, March). Discrimination of consonant articulation location by tactile stimulation of the forearm. In 2010 IEEE Haptics Symposium (pp. 47-54). IEEE. [\[Link\]](#)



Figure 2. A participant sitting in the upright posture during the experiment and a graphical instruction of factors on the forearm.

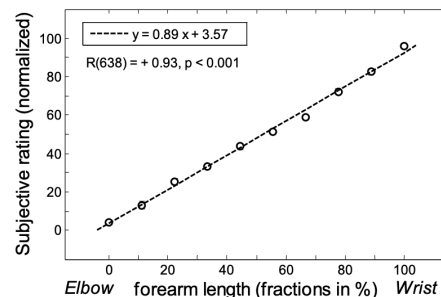


Figure 4. Mean subjective rating (normalized) as a function of the normalized forearm length

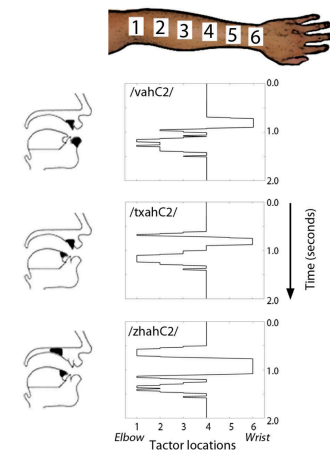


Figure 5. Location of /v/, /t/ and /z/ production inside the mouth and associated tactile cues on the forearm.